SOLUTIONS FOR WORKSHEET 5

Question 1. Prove that the following functions are continuous at the given point using the ε-δ definition of continuity.

(a) $f:\mathbb{R}\to\mathbb{R}$ given by $f(x)=4x-1$ at $x=2$.

(b) $f:\mathbb{R}\to\mathbb{R}$ given by $f(x)=x^2$ at $x=6$.

(c) $f:\mathbb{R}\to\mathbb{R}$ given by $f(x)=x^2-3x$ at $x=2$.

(d) $f:\mathbb{R}\setminus\{-5\}\to\mathbb{R}$ given by $f(x)=\frac{x+2}{x+5}$ at $x=1$.

Proof. (a) We prove that $\lim_{x\to 2}(4x-1)=f(2)=7$. Let $\varepsilon>0$. Consider

$$\big|f(x)-f(2)\big|=\big|4x-1-7\big|=4\big|x-2\big|.$$

Hence, choose $\delta=\varepsilon/4$ so that $\big|x-2\big|<\delta\Rightarrow\big|f(x)-f(2)\big|<\varepsilon$.

(b) We prove that $\lim_{x\to 6}x^2=f(6)=36$. Let $\varepsilon>0$. Consider

$$\big|f(x)-f(6)\big|=\big|x^2-36\big|=\big|(x-6)(x+6)\big|.$$

Now if $\big|x-6\big|<1$, then $-1<x-6<1\Rightarrow 11<x+6<13\Rightarrow\big|x+6\big|<13$. So,

$$\big|x^2-36\big|=\big|(x-6)(x+6)\big|=\big|x+6\big|\big|x-6\big|<13\big|x-6\big|.$$

Set $\delta=\min\{1,\ \varepsilon/13\}$. Then $\big|x-6\big|<\delta\Rightarrow\big|f(x)-L\big|<\varepsilon$.

(c) We prove that $\lim_{x\to 2}(x^2-3x)=-2$. Let $\varepsilon>0$. Consider

$$\big|f(x)-f(2)\big|=\big|x^2-3x+2\big|=\big|(x-2)(x-1)\big|.$$

Now if $\big|x-2\big|<1$, then $-1<x-2<1\Rightarrow 1<x<3\Rightarrow 0<x-1<2\Rightarrow\big|x-1\big|<2$.

Set $\delta=\min\{1,\ \varepsilon/2\}$, then $\big|x-2\big|<\delta\Rightarrow\big|(x-2)(x-1)\big|\le 2\big|x-2\big|\Rightarrow\big|f(x)-f(2)\big|<\varepsilon$.

(d) We prove that $\lim_{x\to 1}\frac{x+2}{x+5}=\frac{1}{2}$. Let $\varepsilon>0$. Consider $\big|f(x)-L\big|=\big|\frac{x+2}{x+5}-\frac{1}{2}\big|=\frac{1}{2}\big|\frac{x-1}{x+5}\big|$. Now if $\big|x-1\big|<1$, then $-1<x-1<1\Rightarrow 5<x+5<7\Rightarrow\frac{1}{7}<\frac{1}{x+5}<\frac{1}{5}\Rightarrow\big|\frac{1}{x+5}\big|<\frac{1}{5}$.

Set $\delta=\min\{1,\ 10\varepsilon\}$, then $\big|x-1\big|<\delta\Rightarrow\big|f(x)-L\big|=\frac{1}{2}\big|\frac{x-1}{x+5}\big|<\frac{1}{10}\big|x-1\big|\Rightarrow\big|f(x)-L\big|<\varepsilon$.

Question 2. For each of the following, give an example or prove no such example exists:

(a) If $\lim_{x\to c}f(x)$ and $\lim_{x\to c}g(x)$ do not exist, can $\lim_{x\to c}\big(f(x)+g(x)\big)$ exist?

(b) If $\lim_{x\to c}f(x)$ and $\lim_{x\to c}g(x)$ do not exist, can $\lim_{x\to c}\big(f(x)g(x)\big)$ exist?

(c) If $\lim_{x\to c}f(x)$ and $\lim_{x\to c}\big(f(x)+g(x)\big)$ exist, must $\lim_{x\to c}g(x)$ exist?

(d) If $\lim_{x\to c}f(x)$ exists and $\lim_{x\to c}\big(f(x)+g(x)\big)$ does not exist, can $\lim_{x\to c}g(x)$ exist?

(e) If $\lim_{x\to c}f(x)$ and $\lim_{x\to c}\big(f(x)g(x)\big)$ exist, must $\lim_{x\to c}g(x)$ exist?

Solution. (a) Take $f(x)=\frac{1}{x-c}$ and $g(x)=-\frac{1}{x-c}$. Then $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ do not exist, but $f(x)+g(x)=0$ and $\lim_{x\to c}\big(f(x)+g(x)\big)$ exist.

(b) Take $f(x)=\begin{cases}1, & x\in\mathbb{Q}\\ 0, & x\notin\mathbb{Q}\end{cases}$ and $g(x)=\begin{cases}0, & x\in\mathbb{Q}\\ 1, & x\notin\mathbb{Q}\end{cases}$. Then $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ do not exist, but $f(x)g(x)=0$ and $\lim_{x\to c}\big(f(x)g(x)\big)$ exist.

(c) The question is equivalent to: If $\lim_{x\to c} h(x)$ and $\lim_{x\to c} f(x)$ exist, must $\lim_{x\to c}\big(h(x)-f(x)\big)$ exist? By Theorem 6.3 (i) and (ii). We know it is true. So $\lim_{x\to c}\big(h(x)-f(x)\big)$ exists and moreover, $\lim_{x\to c}\big(h(x)-f(x)\big)=\lim_{x\to c} h(x)-\lim_{x\to c} f(x)$.

(d) Suppose $\lim_{x\to c} f(x)=L$ and $\lim_{x\to c} g(x)=M$. So, by Theorem 6.3 (ii), $\lim_{x\to c}\big(f(x)+g(x)\big)$ $=L+M$. This contradicts to the given fact that $\lim_{x\to c}\big(f(x)+g(x)\big)$ does not exist. Hence, $\lim_{x\to c} g(x)$ does not exist.

(e) Suppose $\lim_{x\to c} f(x)=L$ and $\lim_{x\to c} h(x)=\lim_{x\to c} f(x)g(x)=M$. So, by Theorem 6.3 (iv)

$$\lim_{x\to c}\frac{h(x)}{f(x)}=\lim_{x\to c} g(x)=\frac{M}{L},\quad L\neq 0$$

provided $f(x)\neq 0$ for all x in its domain. Hence, $\lim_{x\to c} g(x)$ exist under such condition. Otherwise, $\lim_{x\to c} g(x)$ may not exist. For example, $f(x)=0$ and $g(x)$ is Dirichlet's function. Then $\lim_{x\to c} f(x)=0$ exists, $\lim_{x\to c} g(x)$ not exists and $\lim_{x\to c} f(x)g(x)=0$ exists.

Question 3

Let $f:\mathbb{R}\to\mathbb{R}$ be function given by $f(x)=\begin{cases}\sin x, & x\le c\\ ax+b, & x>c\end{cases}$, where $a,b,c\in\mathbb{R}$, $b\ge c$.

For which values of a, if they exist, is f continuous at $x=c$?

Solution. Consider the point $x=c$. To have $f(x)$ continuous at $x=c$, we have

$$\lim_{x\to c^-}\sin x=\sin c=ac+b=\lim_{x\to c^+}\big(ax+b\big)=f(c).$$

Given that $b\ge c$.

Case 1: If $b=c=0$. Then a can be any real number.

Case 2: If $b\ge c\neq 0$. Then $a=\frac{1}{c}\big(\sin c-b\big)$.

Note that $\big|\sin c\big|\le 1\Rightarrow\big|ac+b\big|\le 1\Rightarrow\frac{-1-b}{c}\le a\le\frac{1-b}{c}$.

Case 3: If $0\neq b\ge c=0$. Then $f(x)$ is not continuous at $x=c$.

Because $0=\sin\ c=ac+\ b=\ b\neq 0$, which is impossible.

Question 4. Prove the following limits of functions at the given point directly from the ε-δ definition of a limit.

(a) $\lim\limits_{x\to 0}\left(\frac{1-\sqrt{1-x^2}}{x^2}\right)=\frac{1}{2}$ (b) $\lim\limits_{x\to 1}\left(\frac{x^3}{x-1}-\frac{1}{x-1}\right)=3$ (c) $\lim\limits_{x\to 3}\left(\frac{x^2+2x-15}{x-3}\right)=8$ (d) $\lim\limits_{x\to 2}\left(\frac{x^2-4}{x-2}\right)=4$

Proof. (a) Set $\delta=2\varepsilon<1$. Then $|x|<\delta$ implies that

$$\left|\frac{1-\sqrt{1-x^2}}{x^2}-\frac{1}{2}\right|=\left|\frac{1}{1+\sqrt{1-x^2}}-\frac{1}{2}\right|=\left|\frac{1-\sqrt{1-x^2}}{2\left(1+\sqrt{1-x^2}\right)}\right|=\frac{x^2}{2\left(1+\sqrt{1-x^2}\right)^2}<\frac{1}{2}x^2<\frac{1}{2}|x|<\varepsilon.$$

(b) $\frac{x^3}{x-1}-\frac{1}{x-1}-3=\frac{x^3-1-3x+3}{x-1}=\frac{x^3-3x+2}{x-1}=\frac{(x+2)(x-1)^2}{x-1}=(x+2)(x-1)$.

Suppose $|x-1|<1\Rightarrow 0<x<2\Rightarrow |x+2|<4$. Set $\delta=\min\{1,\frac{\varepsilon}{4}\}$. Then

$$0<|x-1|<\delta\Rightarrow\left|\frac{x^3}{x-1}-\frac{1}{x-1}-3\right|<4\delta=\varepsilon.$$

(c) $\frac{x^2+2x-15}{x-3}-8=\frac{x^2+2x-15-8x+24}{x-3}=\frac{x^2-6x+9}{x-3}=\frac{(x-3)^2}{x-3}=x-3$. Set $\delta=\varepsilon$, then the job is done.

(d) $\frac{x^2-4}{x-2}-4=\frac{x^2-4-4x+8}{x-2}=\frac{x^2-4x+4}{x-2}=\frac{(x-2)^2}{x-2}=x-2$. Set $\delta=\varepsilon$, then the job is done.

Question 5. For each of the following functions, prove directly from the ε-δ definition of continuity that they are continuous on their entire domain.

(a) $f:\mathbb{R}\to\mathbb{R}$ given by $f(x)=|x|$. (b) $g:(0,\infty)\to\mathbb{R}$ given by $g(x)=\sqrt{x}$.

(c) $h:\mathbb{R}\to\mathbb{R}$ given by $h(x)=e^x$. (d) $s:(0,\infty)\to\mathbb{R}$ given by $s(x)=\log x$.

(e) $t:\mathbb{R}\to\mathbb{R}$ given by $t(x)=\cos x$.

Proof. (a) $f(x)=|x|=\begin{cases}x, & x\geq 0\\ -x, & x<0\end{cases}$.

Set $\delta=\varepsilon$. Then $|x-a|<\delta\Rightarrow|f(x)-f(a)|=\big||x|-|a|\big|\leq|x-a|<\delta=\varepsilon$.

See alternative below.

(b) For $a>0$. Set $\delta=\sqrt{a}\varepsilon$, $|x-a|<\delta\Rightarrow|g(x)-g(a)|=\left|\sqrt{x}-\sqrt{a}\right|=\left|\frac{x-a}{\sqrt{x}+\sqrt{a}}\right|<\varepsilon$

because $\sqrt{x}+\sqrt{a}>\sqrt{a}\Rightarrow\frac{1}{\sqrt{x}+\sqrt{a}}<\frac{1}{\sqrt{a}}$. See alternative below.

(c) First, we prove that $h(x)=e^x$ is continuous at $x=0$. Choose $\delta=\ln\varepsilon+1>0$.

Then $|x|<\delta\Rightarrow-\delta<x<\delta\Rightarrow-e^{\delta-1}<e^x-1<e^{\delta-1}\Rightarrow|e^x-1|<e^{\delta-1}=\varepsilon\ \forall x\in\left(-\frac{1}{5},\frac{1}{5}\right)$.

Let $p(x)=x-a$. Then $p(x)$ is continuous at $x=a$. So, by Theorem 4.3, $h\big(p(x)\big)=e^{x-a}$ is continuous at $x=a$. Now $e^x=e^ae^{x-a}$. Therefore, $h(x)=e^x$ is continuous everywhere (by Theorem 4.1). See alternative below.

(d) Prove that $s(x) = \log x = \ln x$ is continuous at $x = 1$.

Use the fact that $e^y > 1 + y$ for all $y > 0$. We have $y > \ln(1 + y)$ for all $y > 0$. Now

$$\left|x - 1\right| < \delta \Rightarrow \left|\ln x - \ln 1\right| = \left|\ln x\right| < \left|x - 1\right| < \delta .$$

Pick $0 < \delta = \varepsilon < 1$ and it is done.

Let $a \in \mathbb{R}^+$ and $f, s : \mathbb{R}^+ \to \mathbb{R}^+$ be functions defined by $f(x) = \frac{x}{a}$ and $s(x) = \log x$. Then f is continuous at $x = a$ and s is continuous at $x = 1$. By Theorem 4.3, it follows that $s\Big(f(x)\Big) = \ln\left(\frac{x}{a}\right)$ is continuous at $x = a$. Since $\ln x = \ln\left(\frac{x}{a}\right) + \ln a$, it follows by Theorem 4.1 that $\ln x$ is continuous everywhere on positive real. See alternative below.

(e) Let $f, g, h : \mathbb{R} \to \mathbb{R}$ be functions defined $f(x) = \sin x$, $g(x) = x - a$ and $h(x) = \cos x$. Then $f \circ g, h \circ g : \mathbb{R} \to \mathbb{R}$ with $f \circ g = \sin(x - a)$ and $h \circ g = \cos(x - a)$. Note that f is continuous at $x = 0$: Pick $\delta = \varepsilon$. Then $\left|x - 0\right| < \delta \Rightarrow \left|\sin x - 0\right| \le \left|x\right| < \delta = \varepsilon$.

h is continuous at $x = 0$: Pick $\delta = \varepsilon$. Then $\left|x - 0\right| < \delta \Rightarrow \left|\cos x - 1\right| \le \left|x\right| < \delta = \varepsilon$.

Note that g is a polynomial and continuous everywhere. It follows by Theorem 4.3 that $f \circ g$ and $h \circ g$ are continuous at $x = a$. Since

$$\cos x = \cos(x - a + a) = \cos(x - a)\cos a - \sin(x - a)\sin a ,$$

it follows by Theorem 4.1 that $t(x) = \cos x$ is continuous everywhere.

Alternative.

Fact 1. Let $f(x) : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x + y) = f(x) + f(y) \quad \forall x, y \in \mathbb{R}$. Then f is continuous at $x = 0$ implies that f is continuous on $\mathbb{R}$.

Proof. Put $x = y = 0$ then $f(0) = 2f(0) \Rightarrow f(0) = 0$. Since $\lim_{x \to 0} f(x) = f(0)$, it follows that $\lim_{x \to a} f(x) = \lim_{h \to 0} f(a + h) = \lim_{h \to 0}\Big(f(a) + f(h)\Big) = f(a) + \lim_{h \to 0} f(h) = f(a) + 0 = f(a)$.

Thus, f is continuous at $x = a$ where $a \in \mathbb{R}$.

Applicable on $f(x) = kx, k \in \mathbb{N}$. Since $f(x + y) = k(x + y) = kx + ky = f(x) + f(y)$ and f is continuous on $x = 0$, it follows that f is continuous on $\mathbb{R}$.

Fact 2. Let $f(x) : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x + y) = f(x)f(y)$ $\forall x, y \in \mathbb{R}$ with $f(0) \neq 0$. Then f is continuous at $x = 0$ implies that f is continuous on $\mathbb{R}$.

Proof. Put $x = y = 0$ then $f(0) = \Big(f(0)\Big)^2 \Rightarrow f(0)\Big(f(0) - 1\Big) = 0 \Rightarrow f(0) = 0, 1$. But given that $f(0) \neq 0$, so $f(0) = 1$. Since $\lim_{x \to 0} f(x) = f(0)$, it follows that

$$\lim_{x \to a} f(x) = \lim_{h \to 0} f(a + h) = \lim_{h \to 0}\Big(f(a)f(h)\Big) = \lim_{h \to 0} f(a) \times \lim_{h \to 0} f(h) = f(a) \times 1 = f(a) .$$

Applicable on $f(x) = e^x$. Since $f(x + y) = e^{x+y} = e^x e^y = f(x)f(y)$, $f(0) = 1 \neq 0$ and f is continuous at $x = 0$, it follows that f is continuous on $\mathbb{R}$.

Fact 3. Let $f(x):\mathbb{R}\to\mathbb{R}$ be a function such that $f(xy)=f(x)+f(y)$. Then f is continuous at $x=1$ implies that f is continuous on $\mathbb{R}$.

Proof. Put $x=y=1$ then $f(1)=2f(1)\Rightarrow f(1)=0$. Since $\lim_{x\to 1} f(x)=f(1)=0$, it follows that $\lim_{x\to a} f(x)=\lim_{h\to 1} f(ah)=\lim_{h\to 1}\Big(f(a)+f(h)\Big)=f(a)+\lim_{h\to 1} f(h)=f(a)+f(1)=f(a)$.

Thus, f is continuous at $x=a$ where $a\in\mathbb{R}$.

Applicable on $f(x)=\ln x, x>0$.

Fact 4. Let $f(x):\mathbb{R}\to\mathbb{R}$ be a function such that $f(xy)=f(x)f(y)$ with $f(1)\neq 0$. Then f is continuous at $x=1$ implies that f is continuous on $\mathbb{R}$.

Proof. Put $x=y=1$ then $f(1)=\Big(f(1)\Big)^2\Rightarrow f(1)=1\because f(1)\neq 0$. Since $\lim_{x\to 1} f(x)=f(1)$, it follows that $\lim_{x\to a} f(x)=\lim_{h\to 1} f(ah)=\lim_{h\to 1}\Big(f(a)f(h)\Big)=f(a)\lim_{h\to 1} f(h)=f(a)f(1)=f(a)$.

Thus, f is continuous at $x=a$ where $a\in\mathbb{R}$.

Applicable on $f(x)=x^r, r\in\mathbb{R}$ ($x>0$ depends on r); and $g(x)=\Big|x\Big|$.

Question 6. Let $f:D\to\mathbb{R}$ be a continuous function at c in D. Prove that $\lambda f:D\to\mathbb{R}$ is continuous at c in D.

Proof. Given that $\forall\varepsilon>0,\exists\delta>0\ni\Big|x-c\Big|<\delta\Rightarrow\Big|f(x)-f(c)\Big|<\frac{\varepsilon}{|\lambda|+1}$. Then

$$\Big|x-c\Big|<\delta\Rightarrow\Big|\lambda f(x)-\lambda f(c)\Big|=\Big|\lambda\Big|\Big|f(x)-f(c)\Big|<\frac{|\lambda|}{|\lambda|+1}\varepsilon<\varepsilon.$$

Question 7. Let $f:\mathbb{R}\to\mathbb{R}$ be a function given by $f(x)=x^{\frac{1}{3}}$.

(a) Prove that f is continuous at 0. (b) Prove that f is continuous on $\mathbb{R}$.

Proof. (a) Let $\varepsilon>0$ be arbitrary and set $\delta=\varepsilon^3$. Then

$$\Big|x-0\Big|<\delta=\varepsilon^3\Rightarrow-\varepsilon^3<x<\varepsilon^3\Rightarrow-\varepsilon<x^{\frac{1}{3}}<\varepsilon\Rightarrow\Big|x^{\frac{1}{3}}-0\Big|<\varepsilon\because(-a)^{\frac{1}{3}}=-a^{\frac{1}{3}}.$$

(b) Use Fact 4 above. We only need to prove that f is continuous at 1. Set $\delta_1=1$. Then $\Big|x-1\Big|<1$ implies that $0<x<2$, which implies that

$$\Big|x^{\frac{2}{3}}+x^{\frac{1}{3}}+1\Big|>1$$

Set $\delta=\min\Big\{1,\varepsilon\Big\}$ and se the fact that $a^3-b^3=\Big(a-b\Big)\Big(a^2+ab+b^2\Big)$.

$$\Big|x^{\frac{1}{3}}-1\Big|=\frac{\Big|x-1\Big|}{\Big|x^{\frac{2}{3}}+x^{\frac{1}{3}}+1\Big|}<\Big|x-1\Big|<\varepsilon.$$

Question 8. Prove that the function $g_a : \mathbb{R} \to \mathbb{R}$ given by $g_a(x) = x + a$ is continuous for all $a \in \mathbb{R}$ for each $x \in \mathbb{R}$.

Proof. Let $\varepsilon = \delta$ and $c \in \mathbb{R}$. Then

$$\big|x - c\big| < \delta \Rightarrow \big|f(x) - f(c)\big| = \big|x + a - c - a\big| = \big|x - c\big| < \delta = \varepsilon\,.$$

Question 9. Prove that the definition of a function $f : D \to \mathbb{R}$ is continuous at a point $c \in D$ is equivalent to $\lim\limits_{x\to c} f(x) = f(c)$.

Proof. $\lim\limits_{x\to c} f(x) = f(c)$ if and only if $\forall\varepsilon > 0\ \exists\delta > 0$ such that

$$\big|f(x) - f(c)\big| < \varepsilon \text{ whenever } 0 < \big|x - c\big| < \delta\,.*$$

f is continuous at c if and only if $\forall\varepsilon > 0\ \exists\delta > 0$ such that

$$\big|f(x) - f(c)\big| < \varepsilon \text{ whenever } \big|x - c\big| < \delta\,.**$$

If $0 \neq \big|x - c\big| > 0$, then * and ** are equivalent. Suppose $\big|x - c\big| = 0$. Then $x = c$. Since $c \in D$, it follows that $\big|f(x) - f(c)\big| = \big|f(c) - f(c)\big| = 0 < \varepsilon$, which is true for both * and **, for all $x \in D$.

Question 10. Let $f, g : D \to \mathbb{R}$ be functions such that $\lim\limits_{x\to c} f(x) = L$ and $\lim\limits_{x\to c} g(x) = M$ for some $c \in \overline{D}$. Prove that

(i) $\lim\limits_{x\to c} \lambda f(x) = L\ \forall\lambda \in \mathbb{R}$;

(ii) $\lim\limits_{x\to c} \big(f(x) + g(x)\big) = L + M$;

(iii) $\lim\limits_{x\to c} \big(f(x)g(x)\big) = LM$; and

(iv) $\lim\limits_{x\to c} \left(\frac{f(x)}{g(x)}\right) = \frac{L}{M}$, providing $g(x) \neq 0\ \forall x \in D, M \neq 0$.

Proof. Given: $\forall\varepsilon_1, \varepsilon_2 > 0, \exists\delta > 0$ ϶ $0 < \big|x - c\big| < \delta \Rightarrow \big|f(x) - L\big| < \varepsilon_1$ and $\big|g(x) - M\big| < \varepsilon_2$.

(i) Set $\varepsilon_1 = \frac{\varepsilon}{|\lambda|+1}$. Then $0 < \big|x - c\big| < \delta$ implies that

$$\big|\lambda f(x) - \lambda L\big| = \big|\lambda\big|\big|f(x) - L\big| < \big|\lambda\big|\varepsilon_1 = \tfrac{|\lambda|}{|\lambda|+1}\varepsilon < \varepsilon\,.$$

(ii) Set $\varepsilon_1 = \varepsilon_2 = \frac{\varepsilon}{2}$. Then $0 < \big|x - c\big| < \delta$ implies that

$$\big|f(x) + g(x) - (L + M)\big| = \big|f(x) - L\big| + \big|g(x) - M\big| < \varepsilon_1 + \varepsilon_2 = \varepsilon\,.$$

(iii) Set $\varepsilon_1 = \frac{1}{2(|M|+1)}\varepsilon$ and $\varepsilon_2 = \frac{1}{2(|L|+\varepsilon_1)}\varepsilon$. Then $0 < \big|x - c\big| < \delta$ implies that

$$\begin{aligned}
\big|f(x)g(x) - LM\big| &= \big|f(x)g(x) - Mf(x) + Mf(x) + LM\big| \\
&\le \big|f(x)\big|\big|g(x) - M\big| + \big|M\big|\big|f(x) - L\big| = \big|f(x) - L + L\big|\big|g(x) - M\big| + \big|M\big|\big|f(x) - L\big| \\
&\le \Big(\big|f(x) - L\big| + \big|L\big|\Big)\big|g(x) - M\big| + \big|M\big|\big|f(x) - L\big| \\
&< \Big(\varepsilon_1 + \big|L\big|\Big)\varepsilon_2 + \big|M\big|\varepsilon_1 \le \tfrac{1}{2}\varepsilon + \tfrac{|M|}{2(|M|+1)}\varepsilon < \tfrac{1}{2}\varepsilon + \tfrac{1}{2}\varepsilon = \varepsilon
\end{aligned}$$

(iv) Suppose $\lim_{x\to c} g(x) = M > 0$, we first prove that $g(x) > \frac{3}{4}M$. Choose $\varepsilon = \frac{M}{4}$. Then $\left|g(x) - M\right| < \frac{M}{4} \Rightarrow \frac{3M}{4} < g(x) < \frac{M}{4}$. Thus, $\frac{1}{g(x)} < \frac{4}{3M}$. Now set $\varepsilon_2 = \frac{3M^2}{4}\varepsilon$, then $\left|\frac{1}{g(x)} - \frac{1}{M}\right| = \left|\frac{M-g(x)}{Mg(x)}\right| < \frac{4\varepsilon_2}{3M^2} = \varepsilon$. We combine the result in (iii) and $\lim_{x\to c}\frac{1}{g(x)} = \frac{1}{M} > 0$ to conclude that $\lim_{x\to c}\left(\frac{f(x)}{g(x)}\right) = \frac{L}{M}$. When $M < 0$, we add on the result in (i) $\lambda = -1$ to complete the proof.

Question 11. A point $x \in \mathbb{R}$ is a limit point of a set $D \subset \mathbb{R}$ if and only if there exists a sequence $\left(a_n\right)$ with $\lim\left(a_n\right) = x$ and $a_n \in D \setminus \{x\}$ $\forall n \in \mathbb{N}$.

Proof. Note that x is a limit point of D if $\forall \varepsilon > 0, \exists y \in D \setminus \{x\}$ such that $\left|y - x\right| < \varepsilon$.

$\Rightarrow$) Define $\left(a_n\right) = \left(y_1, y_2, y_3, \ldots\right)$, $y_n \in D \setminus \{x\}$ such that $\left|y - x\right| < \varepsilon$ $\forall n \in \mathbb{N}$. Since y_n is arbitrary close to x, it follows that $\lim\left(a_n\right) = x$.

$\Leftarrow$) Suppose there exists a sequence $\left(a_n\right)$ with $\lim\left(a_n\right) = x$. By its definition, we have $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $n \geq N \Rightarrow \left|a_n - x\right| < \varepsilon$. Since $a_n \in D \setminus \{x\}$ $\forall n \in \mathbb{N}$, it follows that x is a limit point.

Question 12. Prove the Boundedness Theorem and the Extreme Value Theorem.

Theorem 8.1. The Boundedness Theorem

Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then f is bounded.

Proof. We prove this by contradiction. Suppose that f is not bounded. This leads to two cases below, either f is unbounded above or unbounded below.

Case 1. f is unbounded above. Then for each $n \geq 1$, there exists $x_n \in [a,b]$ such that

$$f(x_n) > n \qquad (1)$$

(to show this: if such sequence (x_n) does not exist then $f(x_n) \leq n$ for all $x_n \in [a,b]$, which means f is bounded above, a contradiction). Since the sequence (x_n) is bounded, so by the BWT, there exists a convergent subsequence (x_{n_k}) which converges to some limit $L \in [a,b]$. Now f is continuous, we have $\lim_{n_k\to\infty} f(x_{n_k}) = f(L)$ (by Theorem 7.2, Chapter 4), so the sequence $\left(f(x_{n_k})\right)$ is converges, and thus $\left(f(x_{n_k})\right)$ is <u>bounded</u>*. From (1), $f(x_{n_k}) > n_k$. But sequence (n) is a strictly increasing sequence (which is always unbounded). So, the sequence $\left(f(x_{n_k})\right)$ is <u>unbounded</u>**. Statement * and ** leads to a contradiction. Therefore, this concludes the proof.

Case 2. f is unbounded below. The proof is similar to Case 1.

Theorem 8.2. Extreme Value Theorem

Let $f:[a,b]\to\mathbb{R}$ be continuous on $[a,b]$. Then f attains its bounds on $[a,b]$.

Proof. We prove that there exist $c,d\in[a,b]$ such that $f(c)=K_l(f)$ and $f(d)=K_u(f)$; but first and foremost, the supremum and infimum exist because we have proven that f is bounded (and therefore upper and lower bound exist, which implies that the supremum and infimum exist too).

Case I. We show that there $\exists d\in[a,b]$ such that $f(d)=K_u(f)=\sup\{f(x):x\in[a,b]\}$. Since $K_u(f)$ is the supremum, $K_u(f)-\frac{1}{n}$ is not an upper bound for f (in $[a,\ b]$) for all $n\geq 1$. So there exists $x_n\in[a,b]$ (for all n) such that $K_u(f)-\frac{1}{n}<f(x_n)\leq K_u(f)$. By the Squeeze Theorem, $\lim_{n\to\infty}f(x_n)=K_u(f)$ (2). Since the sequence (x_n) is bounded, so by the BWT, there exists a convergent subsequence (x_{n_k}) which converges to some limit $d\in[a,b]$. Next, Theorem 7.2 (Chapter 4) guarantee that $\lim_{n_k\to\infty}f(x_{n_k})=f(d)$ (as f is continuous). Even though we do not know what is d, but from Theorem 8.3, Chapter 2 (if a sequence converges to M then all of its subsequences converge to M too), (2) implies that $f(d)=\lim_{n_k\to\infty}f(x_{n_k})=K_u(f)$.

Case II is similar to Case I. The proof is completed.

Question 13. Consider the following functions $f:\mathbb{R}\to\mathbb{R}$.

(a) f is onetinuous at the point $c\in\mathbb{R}$ if for $\forall\varepsilon>0$, we can choose $\delta=1$ and it follows that $|f(x)-f(c)|<\varepsilon$ whenever $|x-c|<\delta$. Give an example of a function that is onetinuous on $\mathbb{R}$.

(b) f is equaltinuous at point $c\in\mathbb{R}$ if for $\forall\varepsilon>0$, we can choose $\delta=\varepsilon$ and it follows that $|f(x)-f(c)|<\varepsilon$ whenever $|x-c|<\delta$. Given an example of a function that is equaltinuous on $\mathbb{R}$ and nowhere onetinuous; or explain why there is no such function.

(c) f is lesstinuous at point $c\in\mathbb{R}$ if for $\forall\varepsilon>0$, we can choose $0<\delta<\varepsilon$ and it follows that $|f(x)-f(c)|<\varepsilon$ whenever $|x-c|<\delta$. Given an example of a function that is lesstinuous on $\mathbb{R}$ and nowhere equaltinuous; or explain why there is no such function.

(d) Is every lesstinuous function continuous? Is every continuous function lesstinuous? Explain your reasoning in both cases.

Answer. (a) The constant function $f(x) = k$ is the only onetinuous function.

(b) $f(x) = x$ is equaltinuous and nowhere onetinuous.

(c) $f(x) = 2x$ is lesstinuous but nowhere equaltinuous.

(d) Every lesstinuous function is continuous, since the definition of lesstinuous is just continuous plus the requirement that $0 < \delta < \varepsilon$; and every continuous function is lesstinuous since if $\delta > 0$ works we can set $\delta' < \delta$ and $\delta' < \varepsilon$ so that $\left|x - c\right| < \delta' < \delta$ still implies that $\left|f(x) - f(c)\right| < \varepsilon$.

Question 14. Let $D \subset \mathbb{R}$ and let $\delta > 0$. Then $c \in D$ is said to be an isolated point of D if and only if there exists $x \in A \subset \mathbb{R} \ni \left|x - c\right| < \delta$ and $A \cap D = \{c\}$.

(a) Let $f : D \to \mathbb{R}$. Prove that if $c \in D$ is an isolated point, then f is continuous at c.

(b) Hence, prove that every sequence $a : \mathbb{N} \to \mathbb{R}$ is continuous on $\mathbb{N}$.

Proof. (a) When c is isolated, we can set δ small enough that the only $x \in A$ satisfying $\left|x - c\right| < \delta$ is $x = c$. Then clearly $\left|f(x) - f(c)\right| = 0 < \varepsilon$ since $f(x) = f(c)\ \forall \left|x - c\right| < \delta$.

(b) We have proven that if $c \in D$ is an isolated point f is continuous on D. Now suppose $D = \mathbb{N}$. Set $A = \mathbb{R} \setminus \mathbb{N}$. Then clearly there exists $x \in A \ni \left|x - c\right| < \delta$ and $A \cap D = \phi$. This means that $c \in \mathbb{N}$ is an isolated point. Hence, by (a), every sequence is continuous on $\mathbb{N}$.

Question 15. Prove that the floor function, $f : \mathbb{R} \to \mathbb{R}$, given by $f(x) = \left\lfloor x \right\rfloor$ is nowhere continuous on $\mathbb{Z}$ but everywhere continuous on $\mathbb{R} \setminus \mathbb{Z}$.

Proof. Let $c \in \mathbb{Z}$. Then $\lim\limits_{x \to c^-} \left\lfloor x \right\rfloor = c - 1 \neq c = \lim\limits_{x \to c^+} \left\lfloor x \right\rfloor$. So, f is discontinuous on $\mathbb{Z}$.

Now let $d \in \mathbb{R} \setminus \mathbb{Z}$. Then

$$\lim_{x \to d^-} \left\lfloor x \right\rfloor = \lim_{x \to d^+} \left\lfloor x \right\rfloor = \text{ the largest integer less than } d$$

Hence, the limit exists and $\lim\limits_{x \to d} f(x) = \left\lfloor d \right\rfloor = f(d)$. So, f is continuous on $\mathbb{R} \setminus \mathbb{Z}$.

Question 16. Let $f:\mathbb{R}\to\mathbb{R}$ such that there exists a constant $c\in(0,1)$ with $|f(x)-f(y)|<c|x-y|\ \forall x,y\in\mathbb{R}$.

(a) Prove that f is continuous on $\mathbb{R}$.

(b) Pick some point $y_1\in\mathbb{R}$ and construct a sequence by iteration, explicitly $(y_1,f(y_1),f(f(y_1)),...)$. In general, show that if $y_{n+1}=f(y_n)$, then the resulting sequence (y_n) is a Cauchy sequence. Hence, $\lim(y_n)=y$.

(c) Prove that y is a fixed point of f. That is, $f(y)=y$. Additionally, prove it is unique.

(d) Finally, prove that for any $x\in\mathbb{R}$, the sequence $(x,f(x),f(f(x)),...)$ converges to y, defined in part (b).

Proof. (a) Let $\delta=\frac{\varepsilon}{c}$ to get $|f(x)-f(y)|<c|x-y|<\varepsilon$ whenever $|x-y|<\delta$. (This is the general proof; we could make it shorter by letting $\delta=\varepsilon$ since $0<c<1$.)

(b) Let $y_{n+1}=f(y_n)$. We want $|y_n-y_m|<\varepsilon$. Since $0<c<1$ we have

$$|f(y_n)-f(y_m)|=|y_{n+1}-y_{m+1}|\le c|y_n-y_m|\le\cdots\le c^m|y_{n-m+1}-y_1|.$$

Thus, if we can bound $|y_k-y_1|\le M$ for some constant M (which may depend on y_1) then it is done. By choosing m is large enough so that $c^m\le\frac{\varepsilon}{M}$. Thus,

$$\begin{aligned}|y_k-y_1|&\le|y_1-y_2|+|y_2-y_3|+\cdots+|y_{k-1}-y_k|\\&\le|y_1-y_2|+c|y_1-y_2|+\cdots+c^{k-2}|y_1-y_2|\\&=|y_1-y_2|\sum_{i=0}^{k-2}c^i<|y_1-y_2|\sum_{i=0}^{\infty}c^i=\frac{|y_1-y_2|}{1-c}\end{aligned}$$

which is bounded, hence proved.

(c) Since f is continuous, $f(\lim(y_n))=\lim(f(y_n))$ which is just the same as y shifted one element forward, so clearly $\lim(f(y_n))=y$, showing that y is a fixed point. Now, consider two similar sequences $(a_n)\to a$ and $(b_n)\to b$ where $a_1,b_1\in\mathbb{R}$, $a_{n+1}=f(a_n)$, and $b_{n+1}=f(b_n)$. By the Algebraic Limit Theorem $b=a+\lim(b_n-a_n)$. Now note

$$\lim(b_n-a_n)\le\lim(|b_n-a_n|)\le\lim(c^{n-1}|b_1-a_1|)=0.$$

Therefore $a=b$. This implies that regardless of our starting choice of y_1 we will end up at the same fixed point y. In particular, for a given fixed-point z, if we start at $z_1=z$ then clearly $\lim(z_n)=z$ but also $\lim(z_n)=y$ and therefore $z=y$ and y is a unique fixed point.

(d) See (c).

Question 17. Let $f:[0,1]\to\mathbb{R}$ be continuous with $f(0)=f(1)$.

(a) Show that there must exist $x,y\in[0,1]$ satisfying $|x-y|=\frac{1}{2}$ and $f(x)=f(y)$.

(b) Show that $\forall n\in\mathbb{N},\exists x_n,y_n\in[0,1]\ni |x_n-y_n|=\frac{1}{n}$ and $f(x_n)=f(y_n)$.

(c) If $h\in\left(0,\frac{1}{2}\right)$ is not of the form $\frac{1}{n}$, there does not necessarily exist $|x-y|=h$ satisfying $f(x)=f(y)$. Provide an example that illustrates this using $h=\frac{2}{5}$.

Proof. (a) Let $g(x)=f(x)-f(x+\frac{1}{2})$ and note that $g(x)$ is continuous on $[0,\frac{1}{2}]$. Also, note that $g(0)=f(0)-f(\frac{1}{2})=-g(\frac{1}{2})=f(\frac{1}{2})-f(1)$, and therefore we can apply IVT to $g(x)$ on $[0,\frac{1}{2}]$ to conclude that there must be a root of $g(x)$, and therefore

$$f(x)=f(x+\tfrac{1}{2}),\text{ for some }x\in\left[0,\tfrac{1}{2}\right].$$

(b) Let $g(x)=f(x)-f(x+\frac{1}{n})$ and note that $g(x)$ is continuous on $[0,\frac{n-1}{n}]$. Also, note that, and since $g(0)$ and $\sum_{i=1}^{n-1}g\left(\frac{i}{n}\right)$ have opposite sign there must be some natural number $1\le i\le n-1$ where $g\left(\frac{i}{n}\right)$ has opposite sign to $g(0)$, at which point we can apply IVT in a similar fashion to part (a) and find a root of $g(x)$, completing the proof.

(c) Consider

$$f(x)=\begin{cases}-10\left(x-0\right)+0 & 0<x\le\frac{1}{5}\\ 15\left(x-\frac{1}{5}\right)-2 & \frac{1}{5}<x\le\frac{2}{5}\\ -10\left(x-\frac{2}{5}\right)+1 & \frac{2}{5}<x\le\frac{3}{5}\\ 15\left(x-\frac{3}{5}\right)-1 & \frac{3}{5}<x\le\frac{4}{5}\\ -10\left(x-\frac{4}{5}\right)+2 & \frac{4}{5}<x\le 1\end{cases}$$

One can go through the grunt work of verifying that this meets the requirements, but it's easier to just plot the function (see graphic) - the function takes a corner every $\frac{1}{5}$ along x.

